TECHNICAL ENGINEERING SPECIFICATION

GCMS Clinical AI Backend Architecture

Comprehensive REST API, Local AI Integration & Terminology Database Specification

System Name:	GCMS Clinical AI Consultation Backend
Version:	Version 1.2.0 (FastAPI + Ollama Production Reference)
Engineering Scope:	Backend-for-Frontend (BFF), Terminology Services, AI Orchestration
Technology Stack:	Python 3.11+, FastAPI, SQLAlchemy 2.0, PostgreSQL 16, Ollama, Docker
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1. System Architecture Overview

The GCMS Clinical AI Backend is built as an asynchronous Backend-for-Frontend (BFF) leveraging FastAPI and SQLAlchemy 2.0. It acts as a secure mediation layer between the clinician's React frontend, the existing Clinic Management System (CMS), the local Ollama AI inference runtime, and high-performance PostgreSQL terminology databases.

> **[IMPORTANT] Three-Tier Architectural Separation**
>
> Layer 1 (AI Reasoning): Ollama synthesizes unstructured consultation context into structured candidate suggestions. Layer 2 (Terminology Grounding): PostgreSQL FTS & Trigram engines verify SNOMED CT and SDD codes. Layer 3 (Deterministic Safety): Independent deterministic Python rules verify allergies, interactions, and contraindications.

2. REST API Specification (The 4 Core Services)

Endpoint Route	Method	Request Payload	Responsibilities & Safeguards
/api/v1/terminology/snomed/search	GET	q={term}&page;=1&page;_size=20	Full-text (tsvector) + Trigram (pg_trgm) search over licensed SNOMED CT concepts.
/api/v1/terminology/sdd/search	GET	q={term}&page;=1&page;_size=20	Multi-field search across brand, generic, strength, HSA registration number.
/api/v1/ai/diagnoses/suggest	POST	{consultation_context, max_suggestions}	Ollama differential diagnoses mapped to verified SNOMED codes. Requires doctor review.
/api/v1/ai/medications/suggest	POST	{selected_diagnosis, consultation_context}	Guideline therapeutics matched to SDD with deterministic allergy & safety check.
/api/v1/health	GET	None	Service readiness probe verifying PostgreSQL and Ollama connections.

3. Local Ollama AI Runtime Integration

- **Connection Pooling** — Persistent httpx.AsyncClient with keepalive connections and configurable timeouts (default 120s).
- **Resilience & Retries** — Tenacity exponential backoff retries on transient connection dropouts.
- **Strict JSON Mode** — Forces Ollama to emit structured JSON complying with Pydantic schemas, eliminating parsing failures.
- **Zero Hallucinations** — The AI system prompts explicitly forbid inventing SNOMED CT or SDD codes. Terminology mapping is exclusively performed by the database ground-truth engine.

4. Database Architecture & PostgreSQL Schemas

Table Name	Primary Key	Search Indexes	Entity Responsibility
snomed_concepts	concept_id (VARCHAR 20)	B-Tree (active)	Core SNOMED CT concept registry.
snomed_descriptions	description_id (VARCHAR 20)	GIN (search_vector), GIN (term trgm)	FSN, preferred terms, and synonyms.
sdd_medications	medication_id (VARCHAR 50)	GIN (search_vector), GIN (name/generic)	HSA-registered Singapore medications.
sdd_active_ingredients	id (UUID)	GIN (ingredient_name trgm)	Active chemical ingredients & strengths.
audit_events	id (UUID)	B-Tree (request_id, created_at)	Immutable audit trail (zero PHI).

5. Privacy Boundary & Deterministic Safety Engine

- **Singapore NRIC / FIN Redaction** — Regex pattern matches and redacts national identifiers [STFGM]\d{7}[A-Z] -> [NRIC_REDACTED].
- **Contact Information Scrubber** — Singapore phone numbers (+65 6/8/9xxx xxxx) and email addresses are sanitized.
- **Allergy Cross-Reactivity** — Evaluates patient allergies against cross-reactive families (Penicillins, Cephalosporins, NSAIDs, Sulfas).
- **Drug Interactions & Contraindications** — Evaluates high-risk drug pairs (Warfarin + NSAIDs, Simvastatin + Macrolides) and pregnancy teratogens.